

Winning Space Race with Data Science

Moacir Leandro Camillo

24 de abril de 2025



Contorno

- Resumo
- Introdução
- Metodologia
- Resultados
- Conclusão
- Apêndice

Resumo

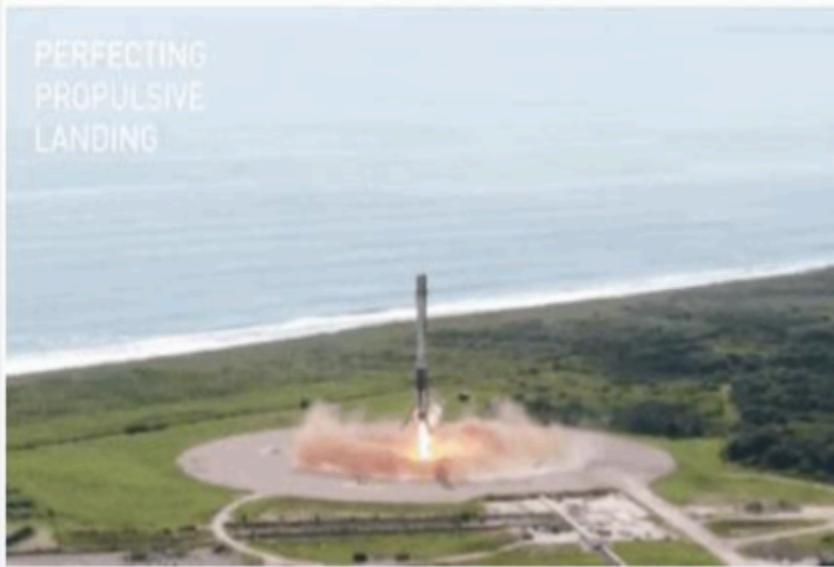
- Resumo das metodologias

- Coleta de dados da SpaceX usando a API da SpaceX
- Coleta de dados da SpaceX com Web Scraping
- Organização de dados da SpaceX
- Análise Exploratória de Dados da SpaceX usando SQL
- Space-X EDA DataViz usando Python Pandas e Matplotlib
- Análise de locais de lançamento da Space-X com análise visual interativa do Folium e Plotly Dash
- Previsão de pouso do SpaceX Machine Learning

- Resumo de todos os resultados

- Resultados da EDA
- Análise visual interativa e painéis
- Análise Preditiva (Classificação)

Introdução



- Histórico e contexto do projeto

A SpaceX anuncia lançamentos de foguetes Falcon 9 em seu site com um custo de 62 milhões de dólares; outros provedores custam mais de 165 milhões de dólares cada, grande parte da economia é porque a SpaceX pode reutilizar o primeiro estágio. Portanto, se pudermos determinar se o primeiro estágio vai pousar, podemos determinar o custo de um lançamento. Essas informações podem ser usadas se uma empresa alternativa quiser fazer uma oferta contra a SpaceX para o lançamento de um foguete.

- Problemas que você deseja encontrar respostas

Neste ponto final, preveremos se o primeiro estágio do Falcon 9 pousará com sucesso usando dados dos lançamentos de foguetes Falcon 9 anunciados em seu site.

Secção 1

Methodology



Metodologia

Resumo

- Metodologia de coleta de dados:
 - Descreve como os conjuntos de dados foram coletados
- Realizar organização de dados
 - Descreve como os dados foram processados
- Realizar análise exploratória de dados (EDA) usando visualização e SQL
- Realizar análises visuais interativas usando Folium e Plotly Dash
- Realizar análises preditivas usando modelos de classificação
 - Como construir, ajustar, avaliar modelos de classificação

Recolha de dados

- Descrição de como os dados do SpaceX Falcon9 foram coletados.
 - Os dados foram coletados pela primeira vez usando a API da SpaceX (uma API RESTful), fazendo uma solicitação get para a API da SpaceX. Isso foi feito primeiro definindo uma série de funções auxiliares que ajudariam no uso da API para extrair informações usando números de identificação nos dados de lançamento e, em seguida, solicitando dados de lançamento de foguetes do URL da API da SpaceX.
 - Finalmente, para tornar os resultados JSON solicitados mais consistentes, os dados de lançamento da SpaceX foram solicitados e analisados usando a solicitação GET e, em seguida, decodificaram o conteúdo da resposta como um resultado Json, que foi então convertido em um quadro de dados do Pandas.
 - Também realizou web scraping para coletar registros históricos de lançamento do Falcon 9 de uma página da [Wikipedia intitulada Lista de lançamentos do Falcon 9](#) e Falcon Heavy dos registros de lançamento são armazenados em aHTML. Usando o BeautifulSoup e as bibliotecas de solicitação, extraio os registros da tabela HTML de inicialização do Falcon 9 da página da Wikipedia, analisei a tabela e a converti em um quadro de dados do Pandas

Coleta de dados – API da SpaceX

- Dados coletados usando a API da SpaceX (API RESTful) fazendo uma solicitação get para a API da SpaceX, em seguida, solicitaram e analisaram os dados de lançamento da SpaceX usando a solicitação GET e decodificaram o conteúdo da resposta como um resultado Json que foi então convertido em um quadro de dados Pandas

- Aqui está a URL do GitHub das chamadas de API da SpaceX concluídas notebook(<https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/1.%20Space-X%20Data%20Collection%20API.ipynb>)

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
static_json_url = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successful with the 200 status response code

```
response.status_code
```

Coloque seu fluxograma de chamadas de API da SpaceX aqui

Now we decode the response content as a json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize method to convert the json result into a dataframe
```

```
resjson = response.json()
data = pd.json_normalize(resjson)
```

Coleta de dados - WebScraping

- Realizou web scraping para coletar registros históricos de lançamento do Falcon 9 de uma Wikipedia usando BeautifulSoup e solicitou, para extrair os registros de lançamento do Falcon 9 da tabela HTML da página da Wikipedia e, em seguida, criou um quadro de dados analisando o launchHTML.

- [Aqui](#) está o URL do GitHub doraspagem da web concluídonotebook.

- (<https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/2.%20Space-X%20Web%20scraping%20Falcon%209%20and%20Falcon%20Heavy%20Launches%20Records%20from%20Wikipedia.ipynb>)

```
In [4]: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"

Next, request the HTML page from the above URL and get a response object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

In [5]: # use requests.get() method with the provided static_url
# assign the response to a object
response = requests.get(static_url)

Create a BeautifulSoup object from the HTML response

In [6]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
soup = BeautifulSoup(response.content, "html.parser")

Print the page title to verify if the BeautifulSoup object was created properly

In [7]: # Use soup.title attribute
soup.title

Out[7]: List of Falcon 9 and Falcon Heavy launches - Wikipedia

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about BeautifulSoup, please check the external re
this lab

In [10]: # use the find_all function in the BeautifulSoup object, with element type 'table'
# Assign the result to a list called 'html_tables'
```

Coloque seu fluxograma de web scraping aqui

Organização de datas

- Depois de obter e criar um Pandas DF a partir dos dados coletados, os dados foram filtrados usando a coluna BoosterVersion para manter apenas os lançamentos do Falcon 9 e, em seguida, lidaram com os valores de dados ausentes nas colunas LandingPad e PayloadMass. Para o PayloadMass, os valores de dados ausentes foram substituídos usando meanvalue of column.

- Também realizei algumas análises exploratórias de dados (EDA) para encontrar alguns padrões nos dados e determinar qual seria o rótulo para treinar modelos supervisionados

- [Aqui](#) está a URL do GitHub dos notebooks relacionados ao datawrangling concluídos.

TASK 1: Create a landing outcome label from Outcome column

Using the Outcome, create a list where the element is zero if the corresponding row in Outcome is variable landing_class:

```
# landing_class = 0 if bad_outcome  
# landing_class = 1 otherwise  
df['Class'] = df['Outcome'].apply(lambda x: 0 if x in bad_outcomes else 1)  
df['Class'].value_counts()
```

```
0  
1
```

Name: Class, dtype: int64

This variable will represent the classification variable that represents the outcome of each launch. if the first stage landed Successfully

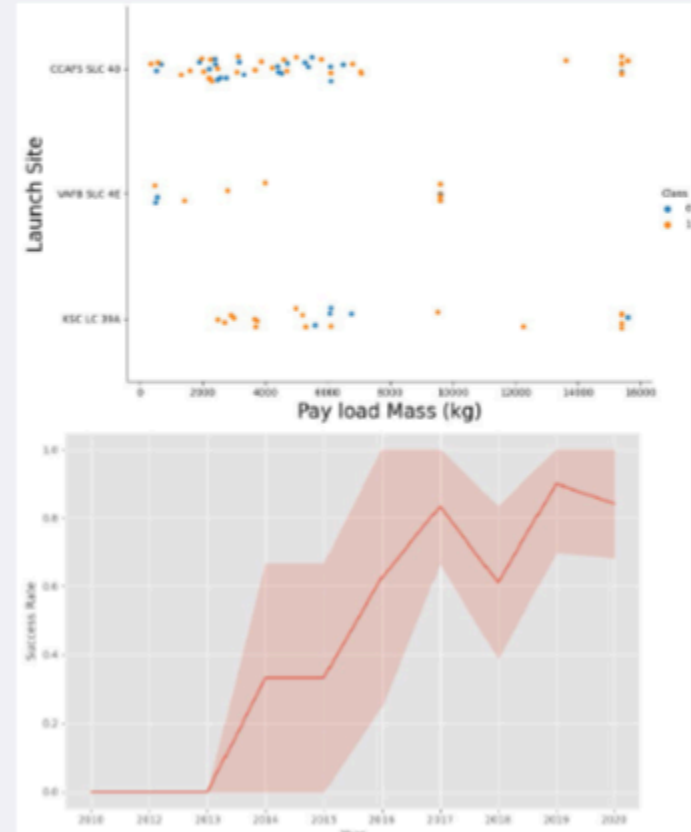
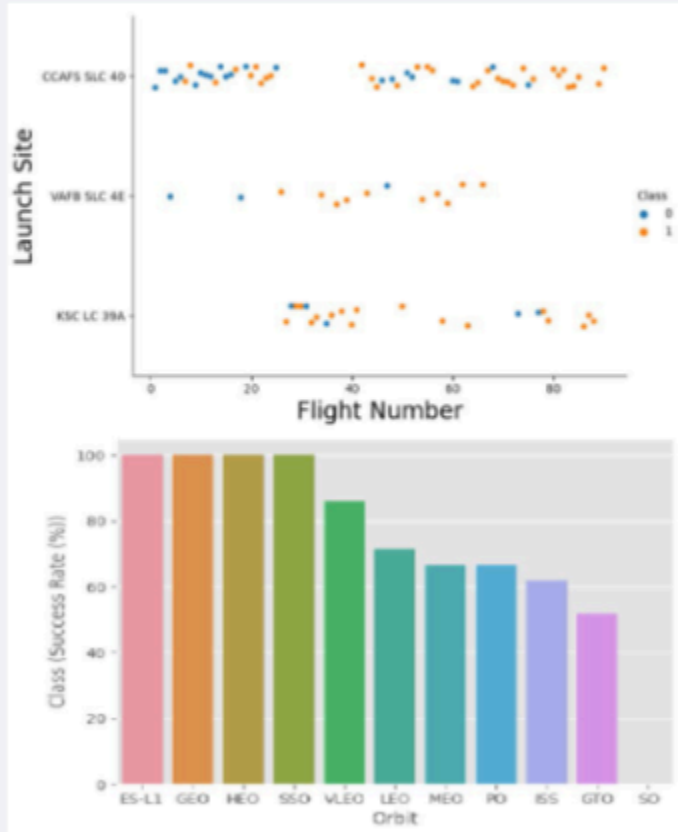
```
landing_class=df['Class']  
df[['Class']].head(8)
```

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

EDA with Data Visualization

- Performed data Analysis and Feature Engineering using Pandas and Matplotlib.i.e.
 - Exploratory Data Analysis
 - Preparing Data Feature Engineering
- Used scatter plots to Visualize the relationship between Flight Number and LaunchSite, Payload and Launch Site, FlightNumber and Orbit type, Payload and Orbittype.
- Used Bar chart to Visualize the relationship between success rate of each orbittype
- Line plot to Visualize the launch success yearly trend.
- [Here is the GitHub URL](https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/5.%20Space-X%20EDA%20DataViz%20Using%20Pandas%20and%20Matplotlib%20-%20SpaceX.ipynb) of your completed EDA with data visualization notebook, (https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/5.%20Space-X%20EDA%20DataViz%20Using%20Pandas%20and%20Matplotlib%20-%20SpaceX.ipynb)

EDA with Data Visualization (Plots Cont....)



EDA with SQL

- The following SQL queries were performed for EDA

- Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
```

- Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

- Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

- Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1';
```

EDA with SQL (Cont....)

- List the date when the first successful landing outcome in ground pad was achieved

```
%sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing_Outcome" = "Success (ground pad)";
```

- List the names of the boosters which have success in drone ship and have payloadmass greater than 4000 but less than 6000 (%sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS __KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;)

- List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

- [Here](https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/4.%20Space-X%20EDA%20Using%20SQL.ipynb) is the GitHub URL of your completed EDA with SQL notebook.
(<https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/4.%20Space-X%20EDA%20Using%20SQL.ipynb>)

Build an Interactive Map with Folium

- Created folium map to marked all the launch sites, and created map objects suchas markers, circles, lines to mark the success or failure of launches for eachlaunch site.
- Created a launch set outcomes (failure=0 or success=1).
- [Here is the GitHub URL](https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/6.Space-X%20Launch%20Sites%20Locations%20Analysis%20with%20Folium-Interactive%20Visual%20Analytics.ipynb) of the completed interactive map with Folium map, as anexternal reference and peer-review purpose (<https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/6.Space-X%20Launch%20Sites%20Locations%20Analysis%20with%20Folium-Interactive%20Visual%20Analytics.ipynb>)

Build a Dashboard with Plotly Dash

- Built an interactive dashboard application with Plotly dash by:
 - Adding a Launch Site Drop-down Input Component
 - Adding a callback function to render success-pie-chart based on selected sitedropdown
 - Adding a Range Slider to Select Payload
 - Addeng a callback function to render the success-payload-scatter-chart scatter plot
- [Here is the GitHub URL](https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/7.%20Build%20an%20Interactive%20Dashboard%20with%20Ploty%20Dash%20-%20spacex_dash_app.py) of your completed Plotly Dash lab(https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/7.%20Build%20an%20Interactive%20Dashboard%20with%20Ploty%20Dash%20-%20spacex_dash_app.py)

SpaceX Launch Records Dashboard

1997

 Springer

Predictive Analysis (Classification)

- Summary of how I built, evaluated, improved, and found the bestperforming classification model
- After loading the data as a Pandas Dataframe, I set out to performexploratory Data Analysis and determine Training Labels by;
 - creating a NumPy array from the column Class in data, by applying the methodto_numpy() then assigned it to the variable Y as the outcome variable.
 - Then standardized the feature dataset (x) by transforming it usingpreprocessing.StandardScaler() function from Sklearn.
 - After which the data was split into training and testing sets using the functiontrain_test_split from sklearn.model_selection with the test_size parameter set to 0.2and random_state to 2.

Predictive Analysis (Classification)

- In order to find the best ML model/ method that would performs best using the test data between SVM, Classification Trees, k nearest neighbors and Logistic Regression;
 - First created an object for each of the algorithms then created a GridSearchCV object and assigned them a set of parameters for each model.
 - For each of the models under evaluation, the GridsearchCV object was created with cv=10, then fit the training data into the GridSearch object for each to Find best Hyperparameter.
 - After fitting the training set, we output GridSearchCV object for each of the models, then displayed the best parameters using the data attribute best_params_ and the accuracy on the validation data using the data attribute best_score_.
 - Finally using the method score to calculate the accuracy on the test data for each model and plotted a confusion matrix for each using the test and predicted outcomes.

Predictive Analysis (Classification)

- The table below shows the test data accuracy score for each of the methods comparing them to show which performed best using the test data between SVM, Classification Trees, k nearest neighbors and LogisticRegression;

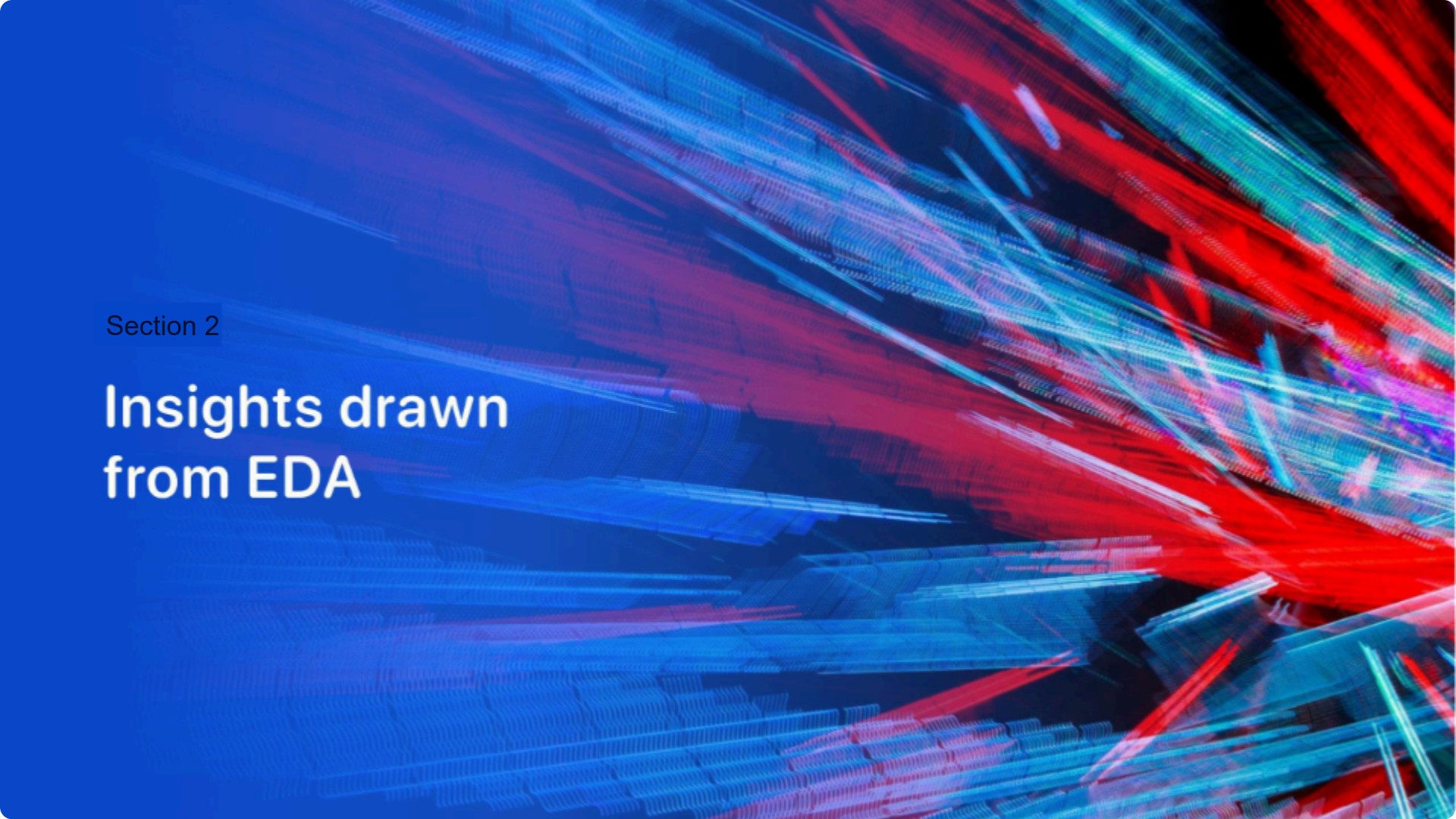
Out[68]:

Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.833333
KNN	0.833333

- GitHub URL of the completed predictive analysis lab(<https://github.com/cgatama/SpaceX-Falcon-9-1st-stage-Success-Landing-Prediction/blob/main/8.%20SpaceX%20Machine%20Learning%20Prediction.ipynb>)

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

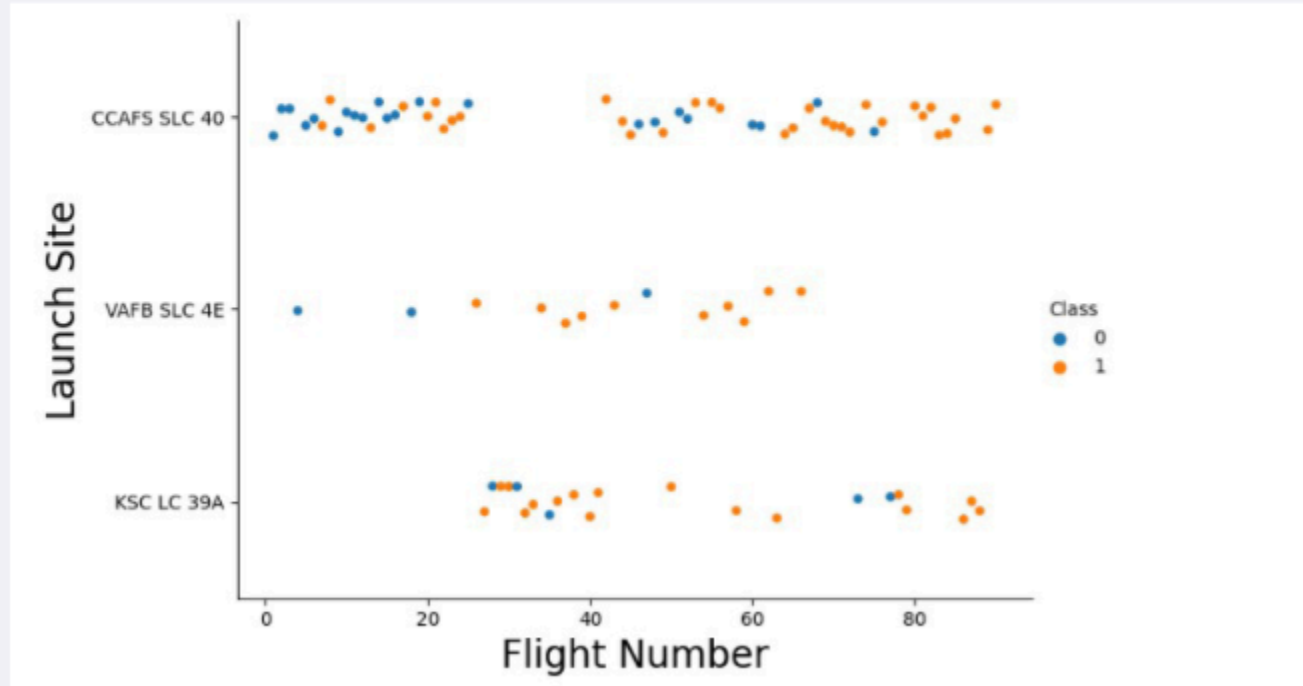
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of bright blue and red streaks and lines on the right. These streaks appear to be digital or data-related, possibly representing a network or a data flow. A faint, light blue grid pattern is visible across the entire background, adding a sense of structure and depth.

Section 2

Insights drawn from EDA

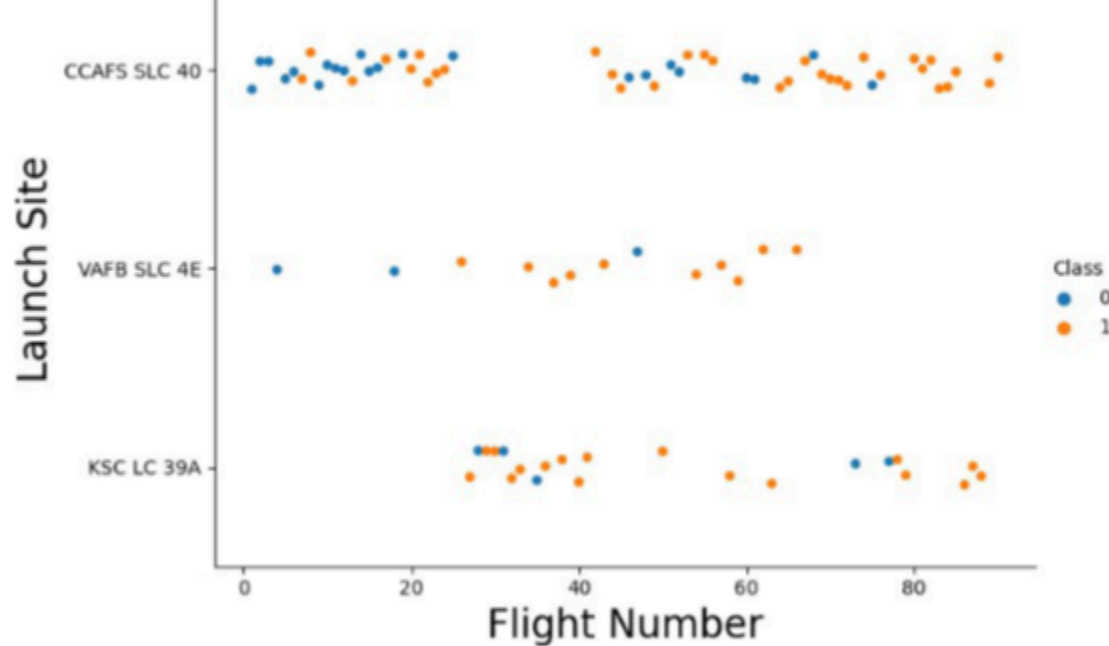
Flight Number vs. Launch Site

A scatter plot of Flight Number vs. Launch Site



Flight Number vs. Launch Site with explanations

A scatter plot with explanations Flight Number vs. Launch Site

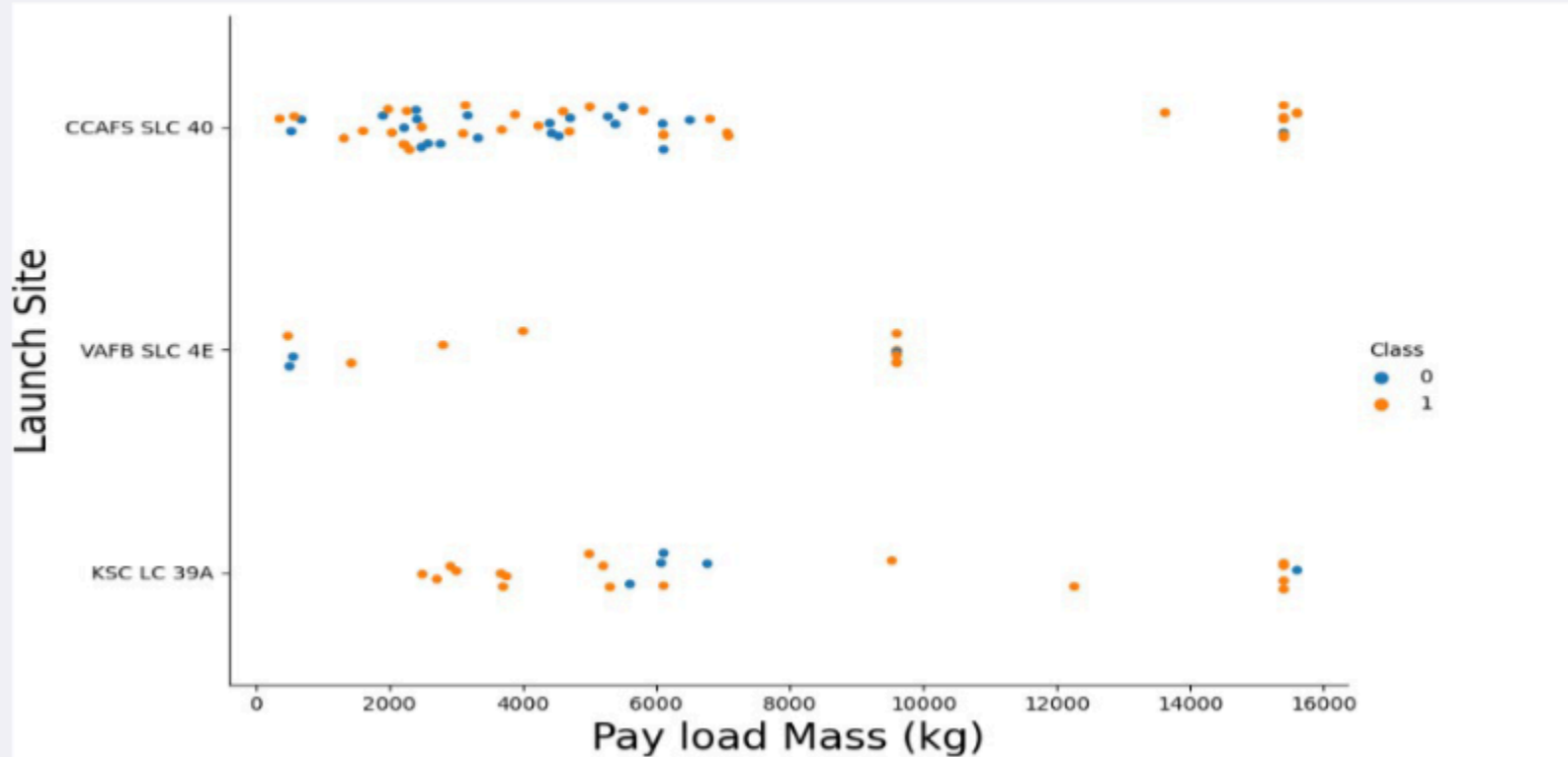


Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

We can deduce that, as the flight number increases in each of the 3 launch sites, so does the success rate. For instance, the success rate for the VAFB SLC 4E launch site is 100% after the Flight number 50. Both KSC LC 39A and CCAFS SLC 40 have a 100% success rates after 80th flight.

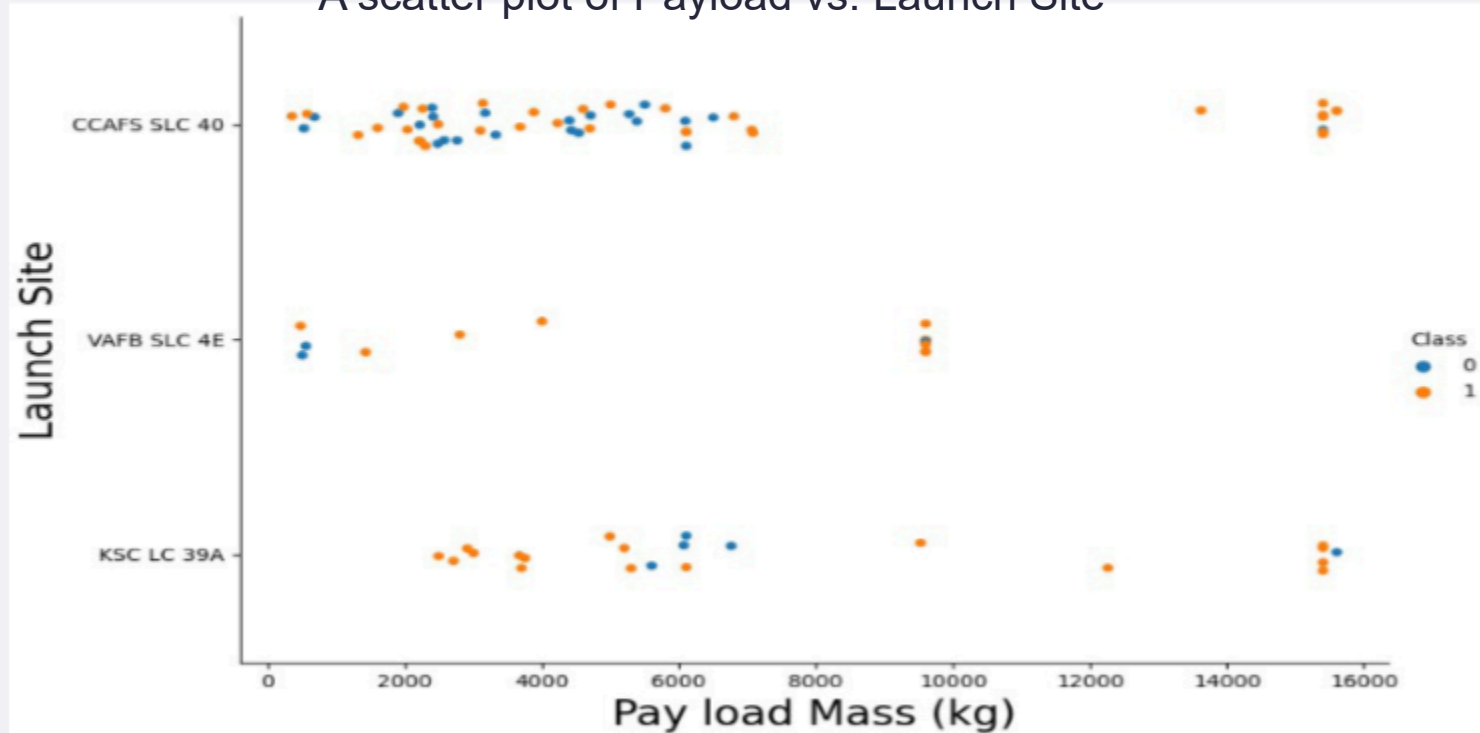
Payload vs. Launch Site

A scatter plot of Payload vs. Launch Site



Payload vs. Launch Site with explanations

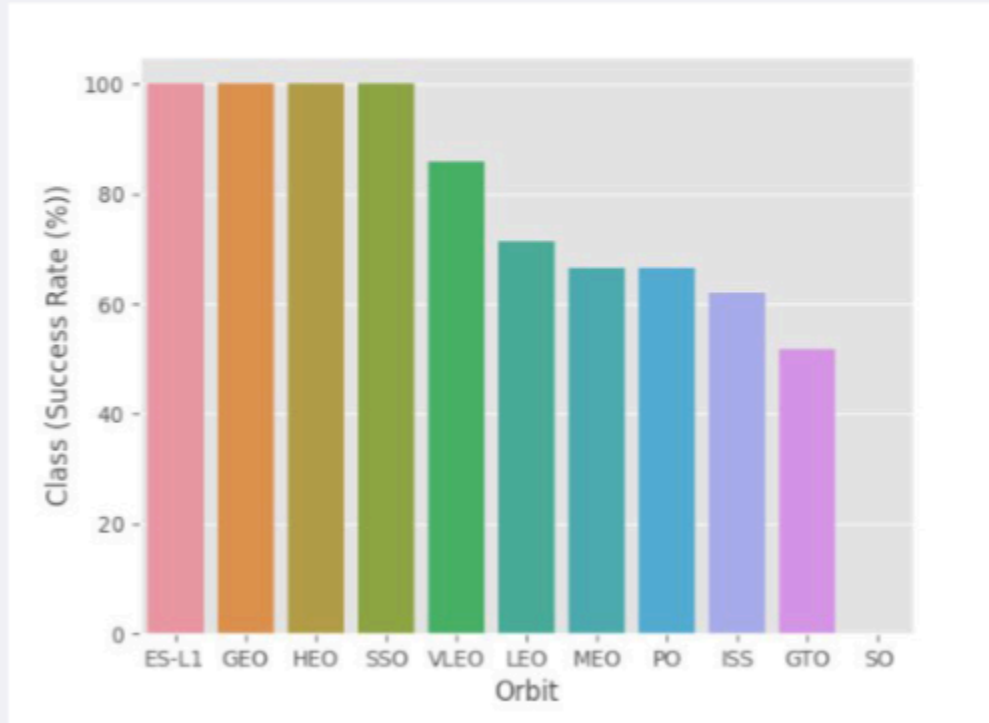
A scatter plot of Payload vs. Launch Site



Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

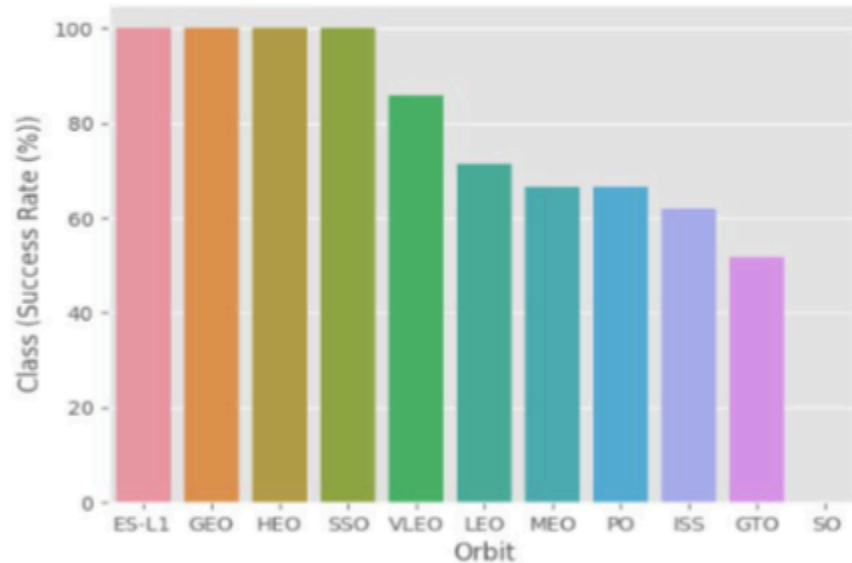
Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type



Success Rate vs. Orbit Type with explanations

- Show the screenshot of the bar chart with explanations

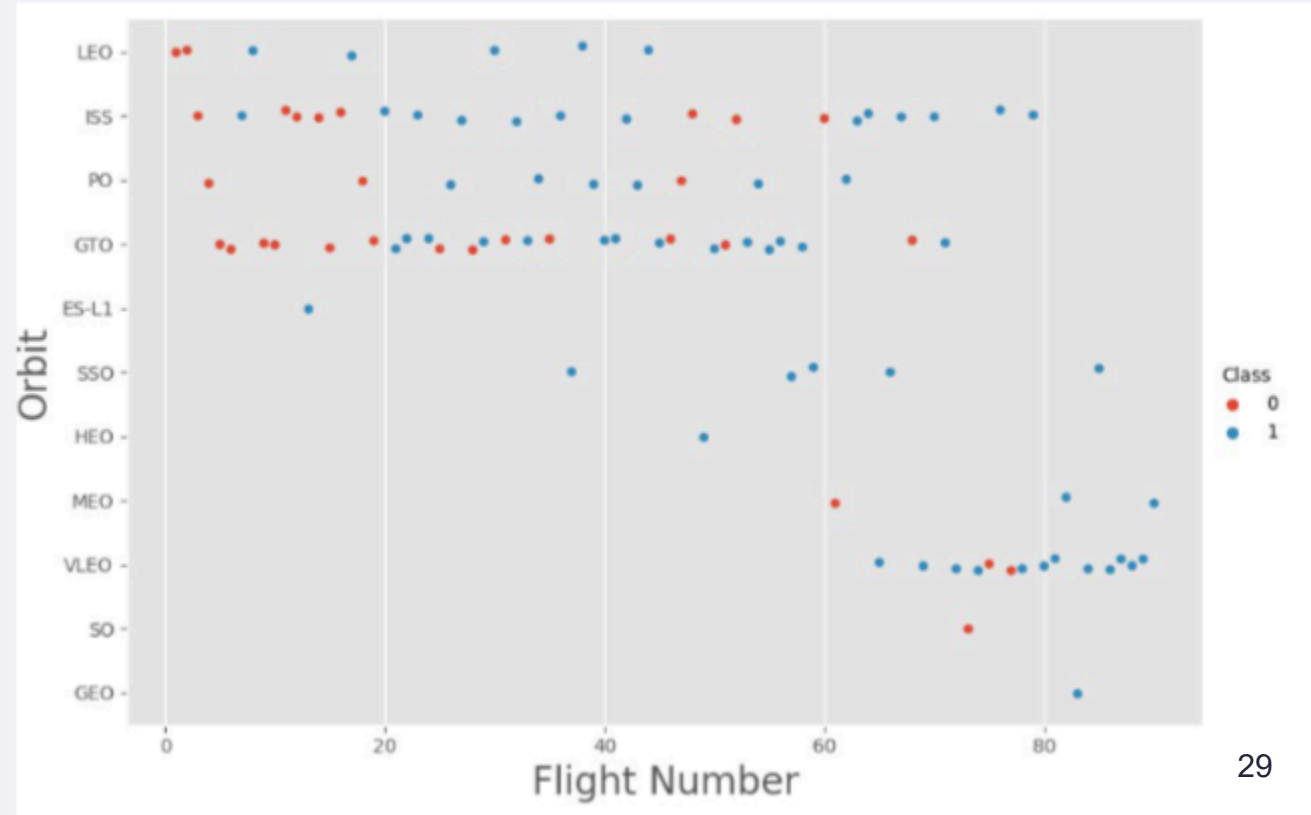


Analyze the plotted bar chart try to find which orbits have high success rate.

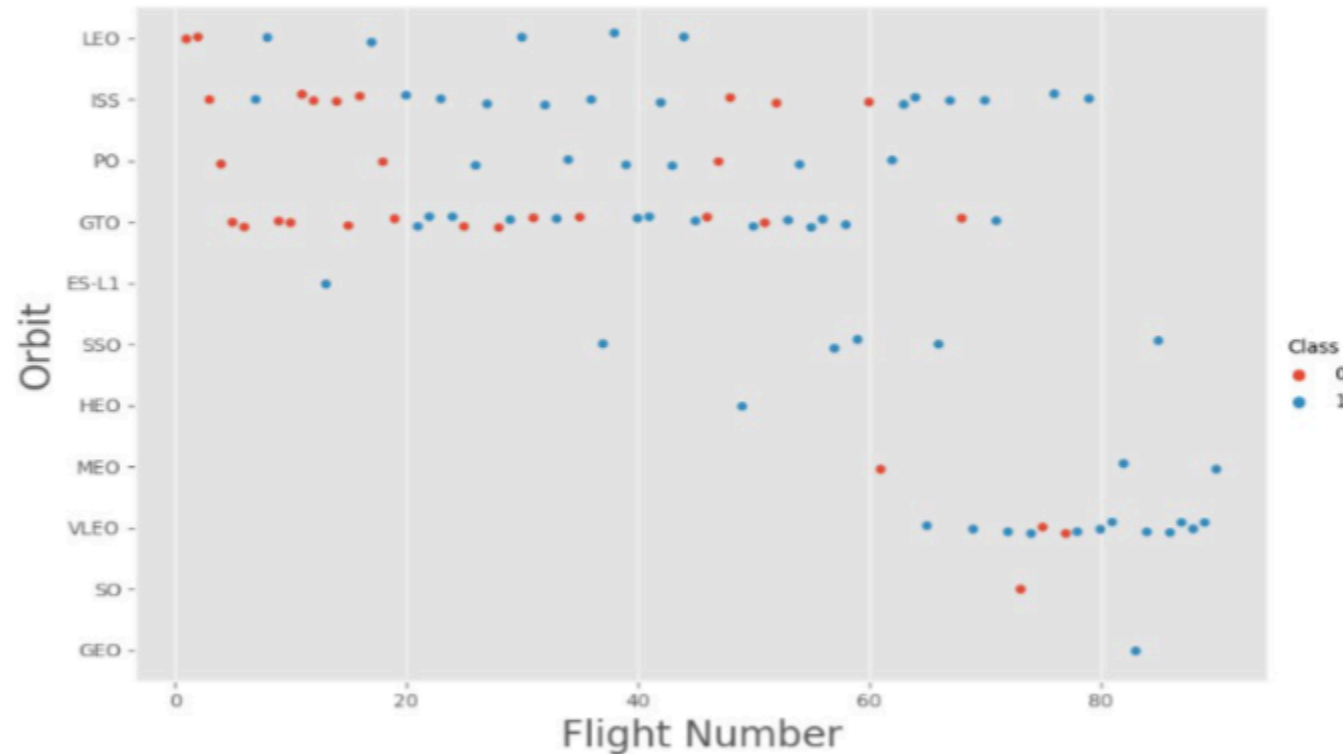
Orbits ES-L1, GEO, HEO & SSO have the highest success rates at 100%, with SO orbit having the lowest success rate at ~50%. Orbit SO has 0% success rate.

Flight Number vs. Orbit Type

- A scatter point of Flight number vs. Orbit type



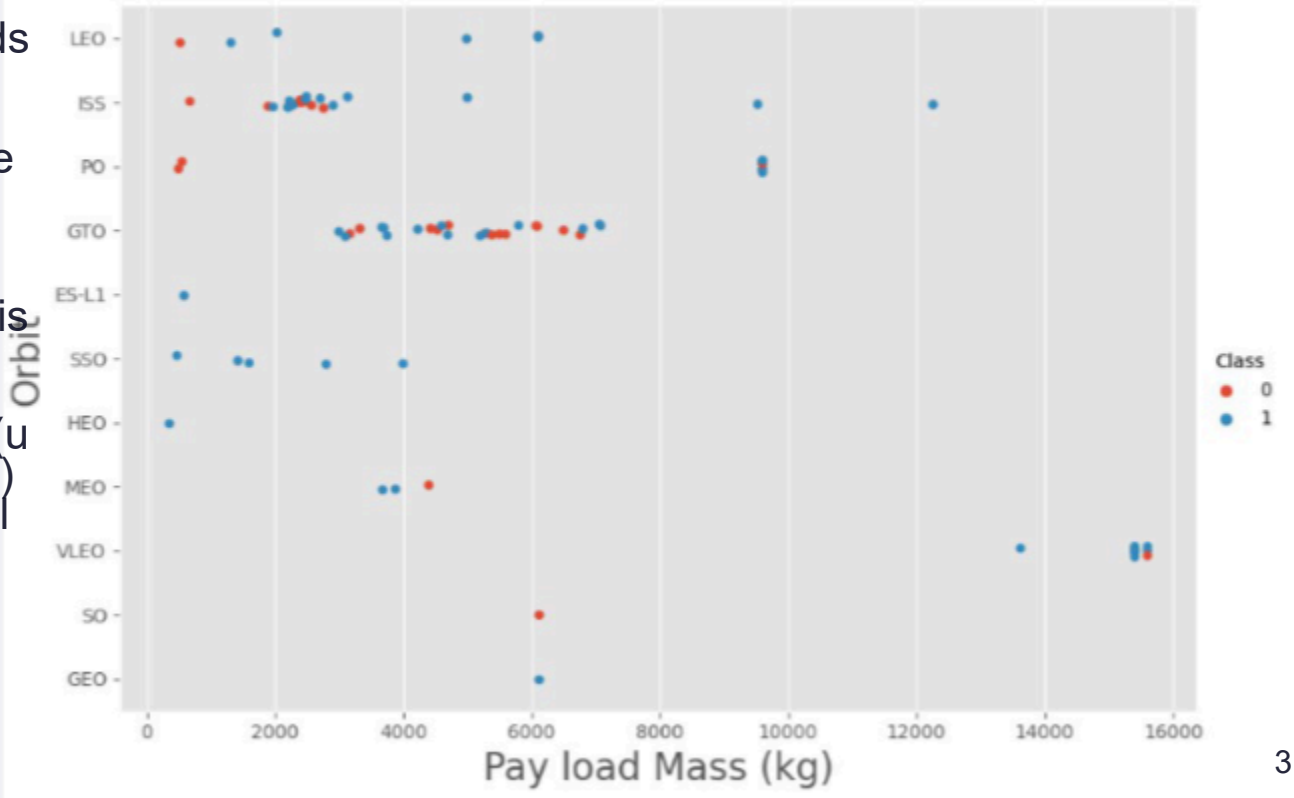
Flight Number vs. Orbit Typewith explanations



You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

Payload vs. Orbit Type

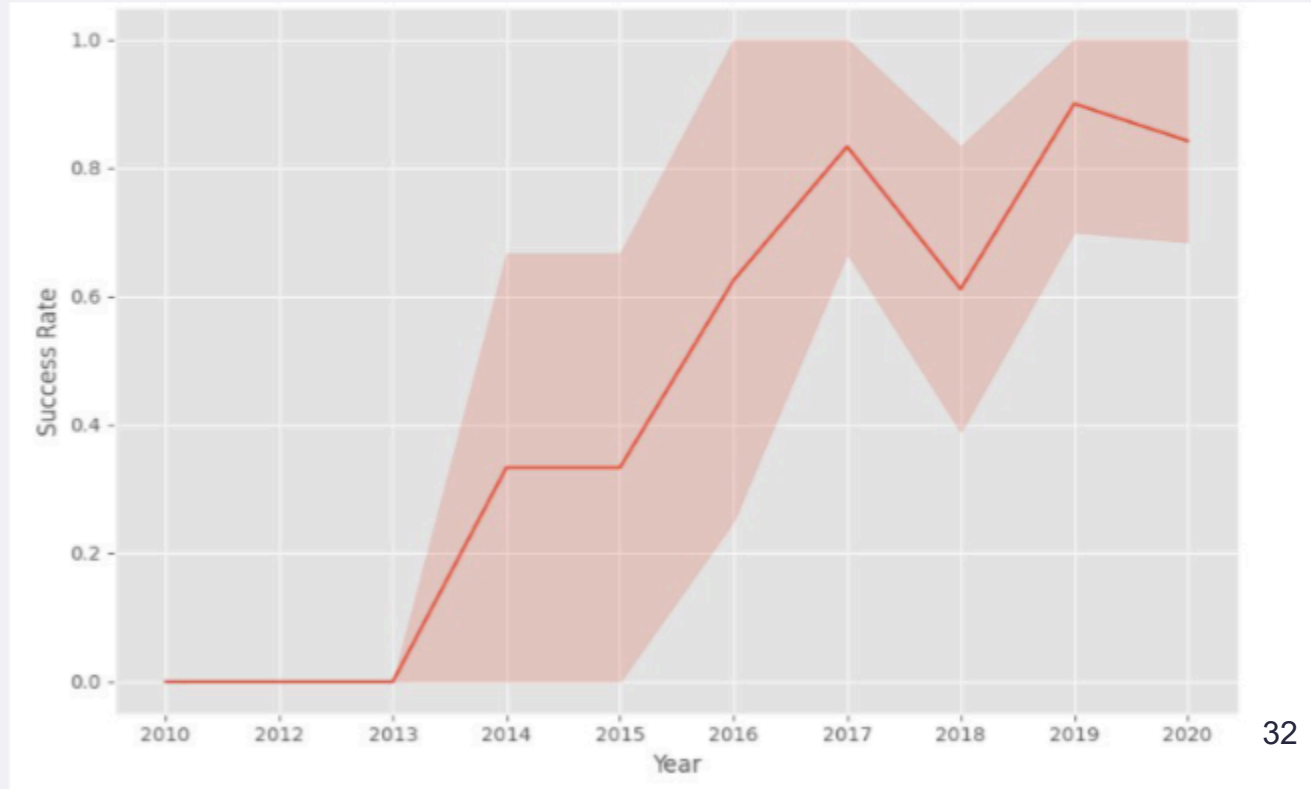
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However for GTO we cannot distinguish this well as both positive landing rate and negative landing (unsuccessful mission) both have near equal chances.



Launch Success Yearly Trend

- Since 2013, the success rate kept going uptill 2020

A line chart of yearly average success rate



All Launch Site Names

- Find the names of the unique launch sites
- Used 'SELECT DISTINCT' statement to return only the unique launch sites from the 'LAUNCH_SITE' column of the SPACEXTBL table

Task 1

Display the names of the unique launch sites in the space mission

```
In [31]: %sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[31]: Launch_Sites
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
In [72]: %sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[72]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
04-06-2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
08-12-2010	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
22-05-2012	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
08-10-2012	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
01-03-2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Used 'LIKE' command with '%' wildcard in 'WHERE' clause to select and display a table of all records where launch sites begin with the string 'CCA'

Total Payload Mass

- Calculate and Display the total payload carried by boosters from NASA

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [17]: %sql SELECT SUM(PAYLOAD_MASS_KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[17]:
```

Total Payload Mass(Kgs)	Customer
45596	NASA (CRS)

- Used the 'SUM()' function to return and display the total sum of 'PAYLOAD_MASS_KG_' column for Customer 'NASA(CRS'

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
* sqlite:///my_data1.db  
Done.
```

Payload Mass Kgs	Customer	Booster_Version
2534.6666666666665	MDA	F9 v1.1 B1003

- Used the 'AVG()' function to return and display the average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing _Outcome" = "Success (ground pad)";
```

```
* sqlite:///my_data1.db  
Done.
```

```
MIN(DATE)
```

```
01-05-2017
```

- Used the 'MIN()' function to return and dispaly the first (oldest) date when firstsuccessful landing outcome on ground pad 'Success (ground pad)'happened.

Successful Drone Ship Landing with Payload between 4000 and 6000

- List of Boosters which have successfully landed on drone ship and had payloadmass greater than 4000 but less than 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
j1: # %sql SELECT * FROM 'SPACEXTBL'

j2: %sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing _Outcome" = "Success (drone ship)" AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000

* sqlite:///my_data1.db
Done.
```

Booster_Version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11 / EchoStar 105

- Used 'Select Distinct' statement to return and list the 'unique' names of boosters with operators >4000 and <6000 to only list booster with payloads between 4000-6000 with landing outcome of 'Success (drone ship)'.

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- Used the 'COUNT()' together with the 'GROUP BY' statement to return total number of missions outcomes

Boosters Carried Maximum Payload

- List of the boosters which have carried the maximum payload mass

```
%sql SELECT "Booster_Version",Payload, "PAYLOAD_MASS_KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTBL)
```

* sqlite:///my_data1.db
Done.

Booster_Version	Payload	PAYLOAD_MASS_KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

- Using a Subquery to return and pass the Max payload and used it list all the boosters that have carried the Max payload of 15600kgs

2015 Launch Records

- List of failed landing outcomes in drone ship, with their booster versions, and launch site names in 2015

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015.

```
%sql SELECT substr(Date,7,4), substr(Date, 4, 2),"Booster_Version", "Launch_Site", Payload, "PAYLOAD_MASS_KG_", "Mission_Outcome", "Landing_Outcome"
```

```
* sqlite:///my_data1.db
```

Done.

substr(Date,7,4)	substr(Date, 4, 2)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Mission_Outcome	Landing_Outcome
2015	01	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	Success	Failure (drone ship)
2015	04	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	Success	Failure (drone ship)

- Used the 'substr()' in the select statement to get the month and year from the date column where substr(Date,7,4)='2015' for year and Landing_outcome was 'Failure (drone ship)' and return the records matching the filter.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success(ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Task 10

Rank the count of successful landing_outcomes between the date 04-06-2010 and 20-03-2017 in descending order.

```
%sql SELECT * FROM SPACEXTBL WHERE "Landing _Outcome" LIKE 'Success%' AND (Date BETWEEN '04-06-2010' AND '20-03-2017') ORDER BY Date DESC;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
19-02-2017	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
18-10-2020	12:25:57	F9 B5 B1051.6	KSC LC-39A	Starlink 13 v1.0, Starlink 14 v1.0	15600	LEO	SpaceX	Success	Success
18-08-2020	14:31:00	F9 B5 B1049.6	CCAFS SLC-40	Starlink 10 v1.0, SkySat-19, -20, -21, SAOCOM 1B	15440	LEO	SpaceX, Planet Labs, PlanetIQ	Success	Success
18-07-2016	04:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
18-04-2018	22:51:00	F9 B4 B1045.1	CCAFS SLC-40	Transiting Exoplanet Survey Satellite (TESS)	362	HEO	NASA (LSP)	Success	Success (drone ship)

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of several vertical panels, giving it a slightly segmented appearance. The left side is a solid blue gradient, while the right side shows the Earth's surface with glowing city lights and cloud patterns.

Section 3

Launch Sites Proximities Analysis

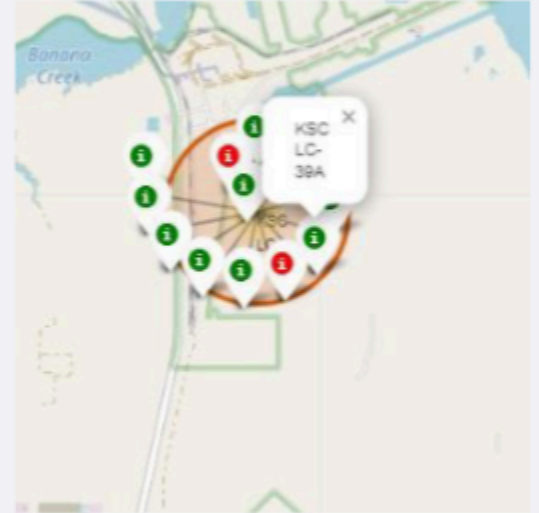
Markers of all launch sites on global map



- All launch sites are in proximity to the Equator, (located southwards of the USmap). Also all the launch sites are in very close proximity to the coast.

Launch outcomes for each site on the map With ColorMarkers

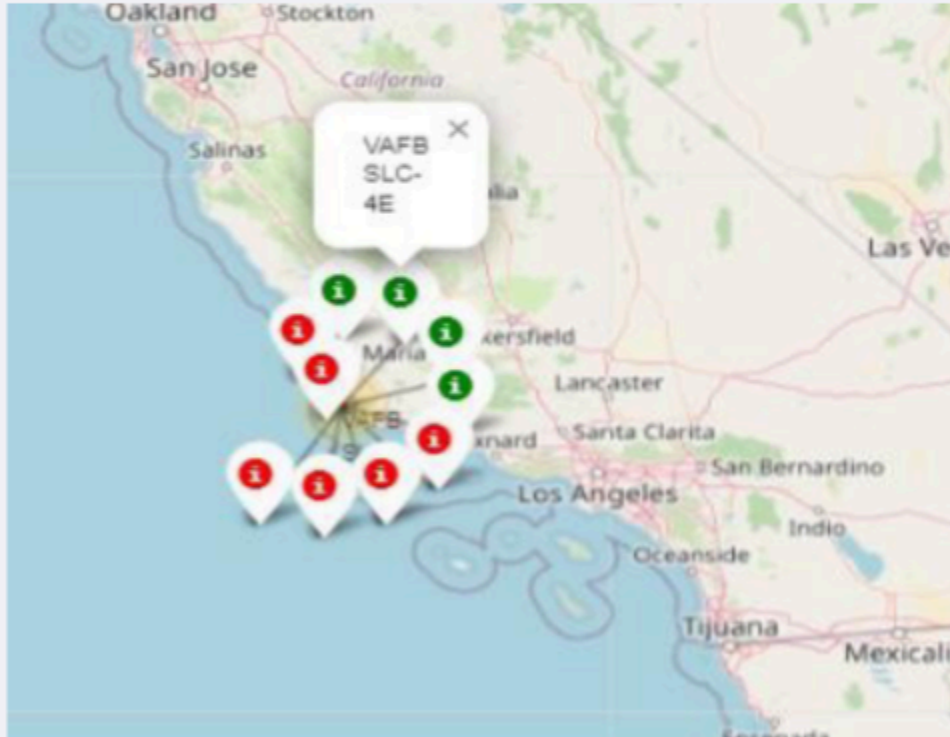
Florida Sites



- In the Eastern coast (Florida) Launch site KSC LC-39A has relatively high successrates compared to CCAFS SLC-40 & CCAFS LC-40.

Launch outcomes for each site on the map With ColorMarkers

West Coast/ Carlifornia



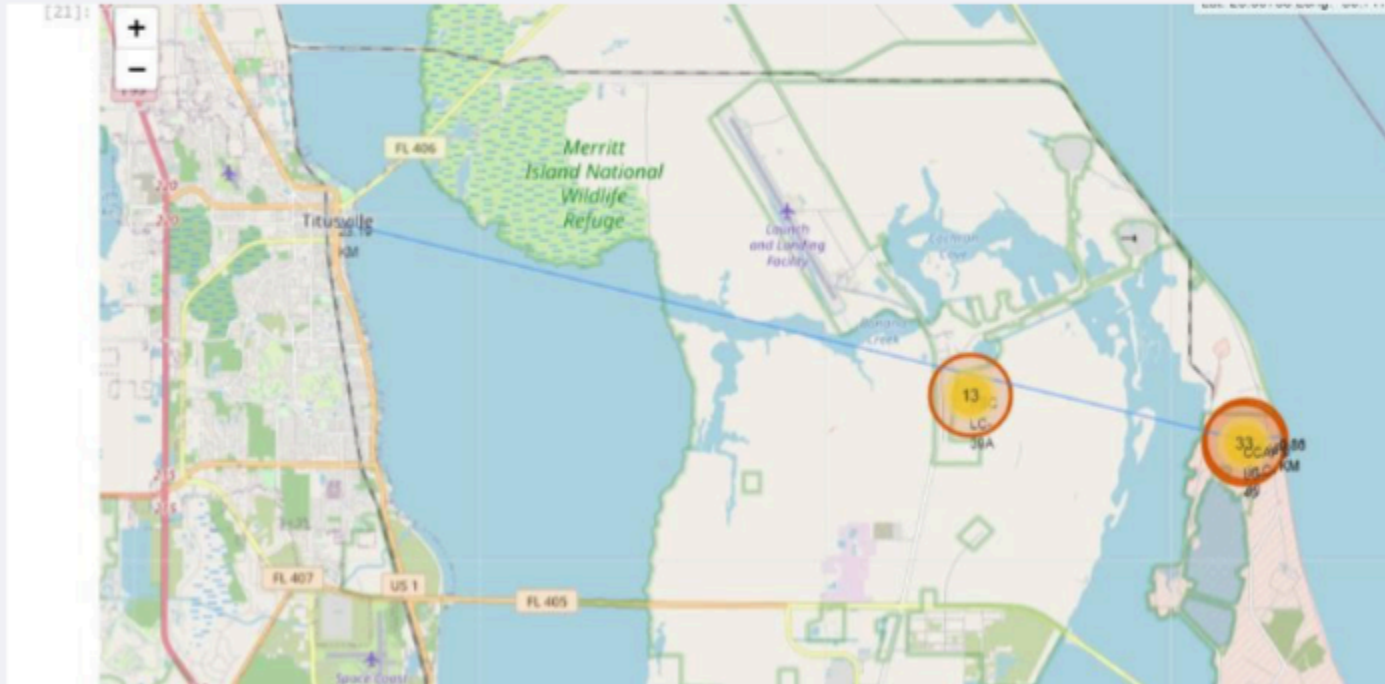
- In the West Coast(California) Launchsite VAFB SLC-4E has relatively lower success rates 4/10 compared to KSC LC-39A launch site in the Eastern Coast of Florida.

Distances between a launch site to its proximities

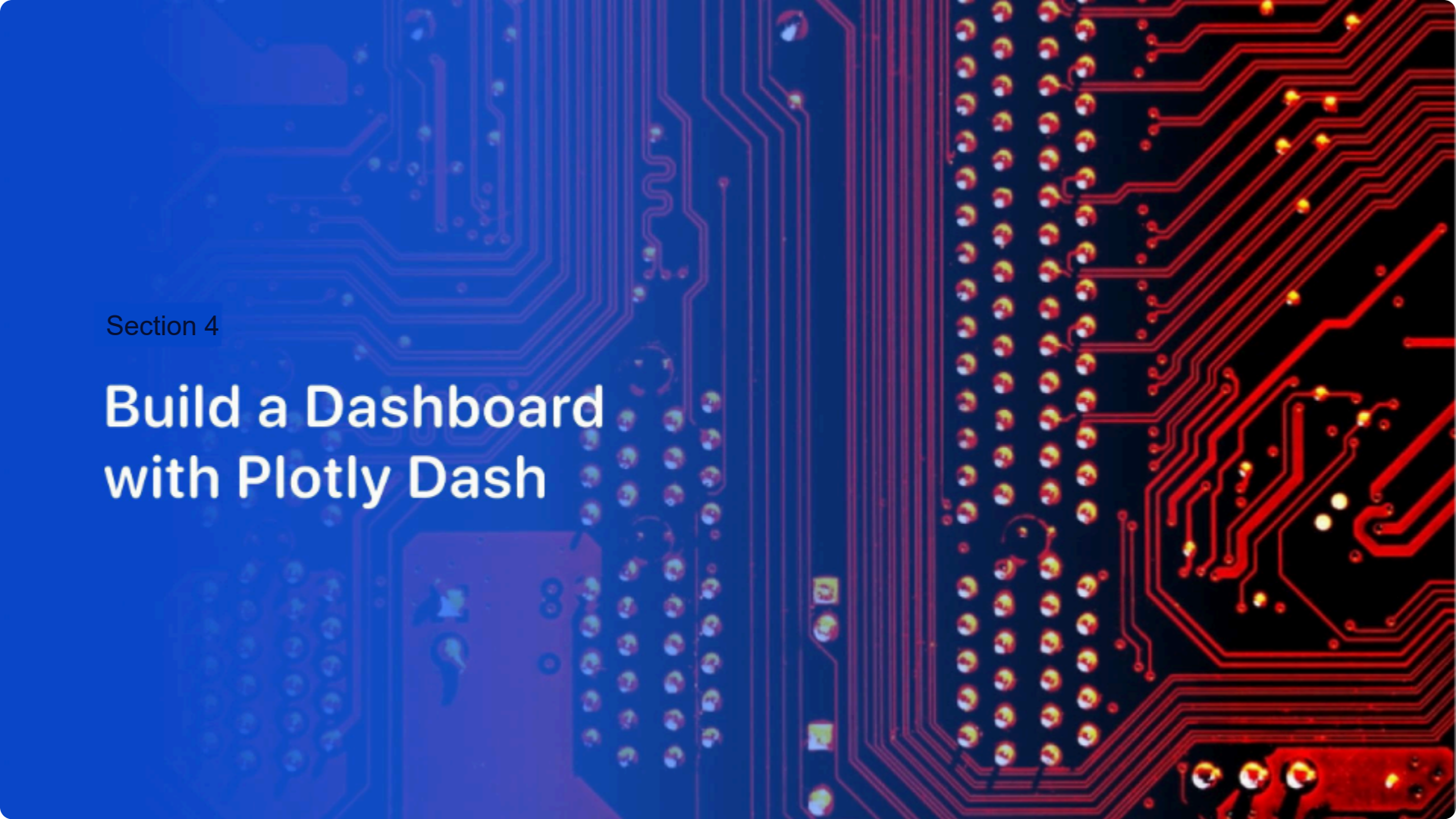


- Launch site CCAFS SLC-40 proximity to coastline is 0.86km

Distances between a launch site to its proximities



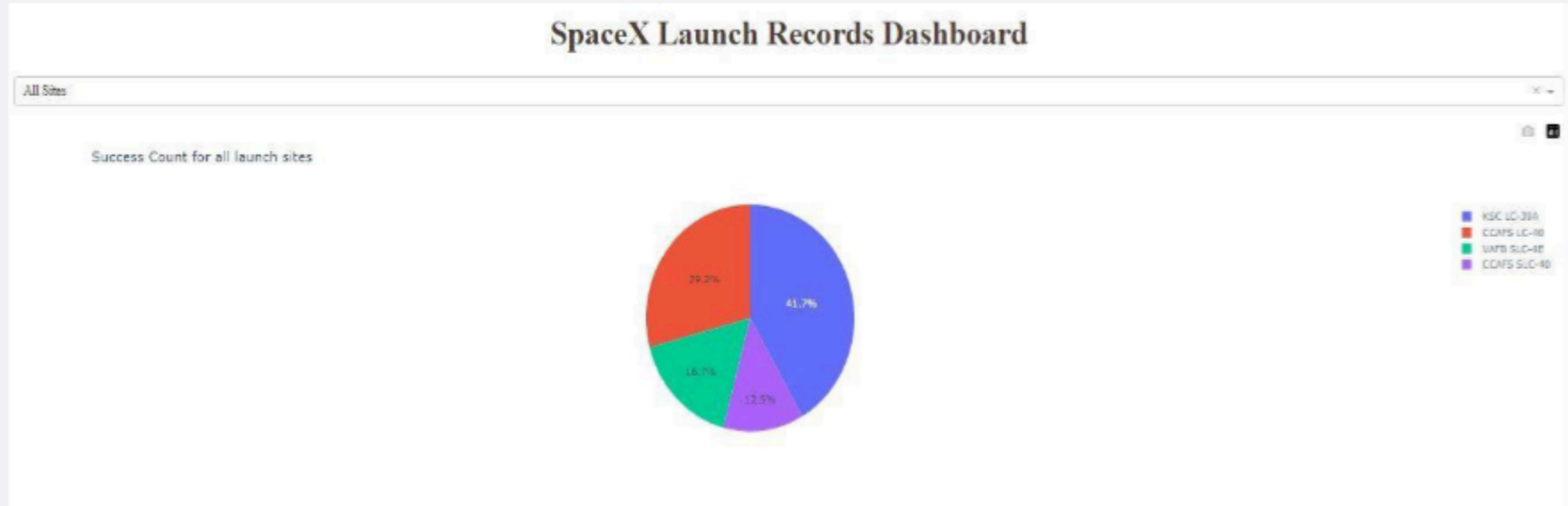
- Launch site CCAFS SLC-40 closest to highway (Washington Avenue) is 23.19km



Section 4

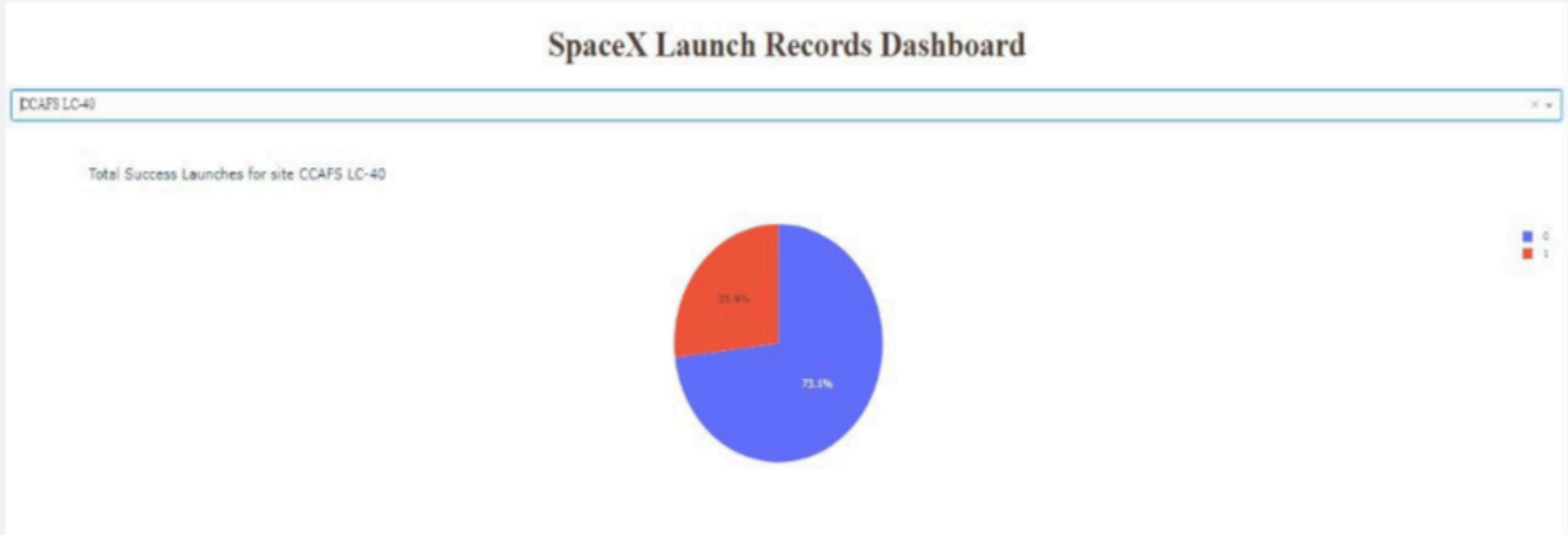
Build a Dashboard with Plotly Dash

Pie-Chart for launch success count for all sites



- Launch site KSC LC-39A has the highest launch success rate at 42% followed by CCAFS LC-40 at 29%, VAFB SLC-4E at 17% and lastly launch site CCAFS SLC-40 with a successrate of 13%

Pie chart for the launch site with 2nd highest launch success ratio



- Launch site CCAFS LC-40 had the 2nd highest success ratio of 73% success against 27% failed launches

Payload vs. Launch Outcome scatter plot for all sites



- For Launch site CCAFS LC-40 the booster version FT has the largest successrate from a payload mass of >2000kg



Section 5

Predictive Analysis (Classification)

Classification Models Accuracy

Out[68]:

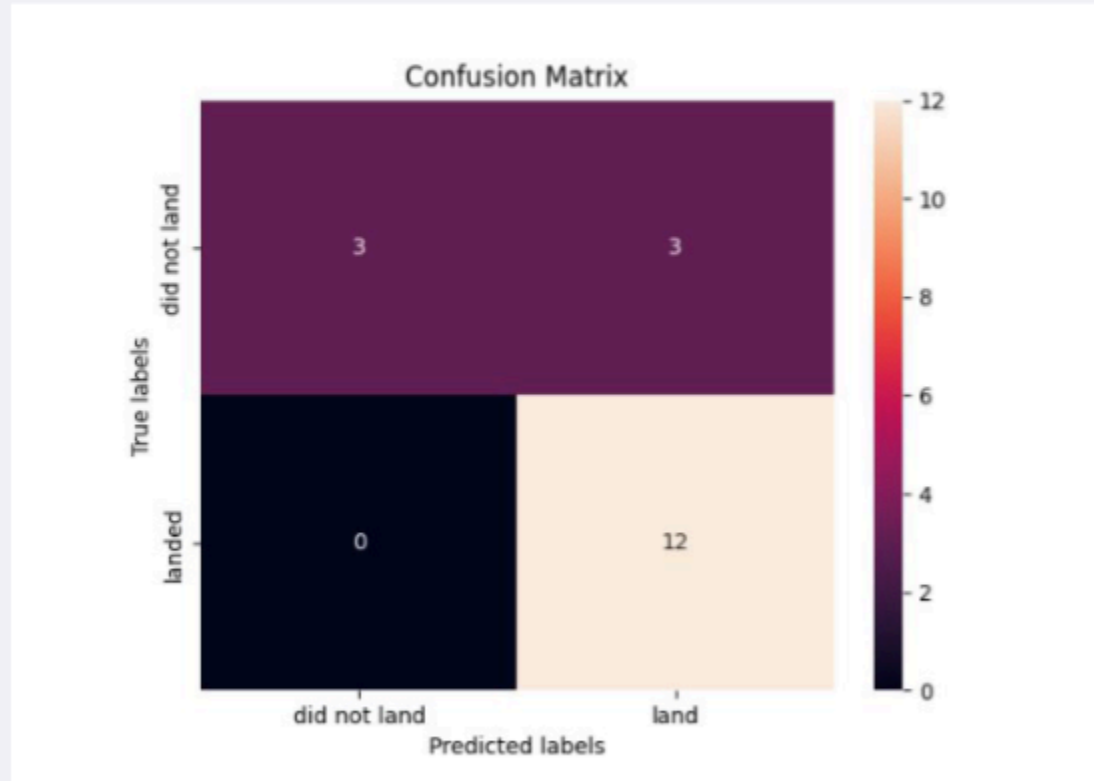
0

Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.833333
KNN	0.833333

All the methods perform equally on the test data: i.e. They all have the same accuracy of 0.833333 on the test Data

Confusion Matrix

- All the 4 classification models had the same confusion matrices and were able to equally distinguish between the different classes. The major problem is false positives for all the models.



Conclusions

- Different launch sites have different success rates. CCAFS LC-40, has a success rate of 60 %,while KSC LC-39A and VAFB SLC 4E has a success rate of 77%.
- We can deduce that, as the flight number increases in each of the 3 launch sites, so does the success rate. For instance, the success rate for the VAFB SLC 4E launch site is 100% after the flight number 50. Both KSC LC 39A and CCAFS SLC 40 have a 100% success rates after 80th flight
- If you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launch site there are no rockets launched for heavy payload mass (greater than 10000).
- Orbits ES-L1, GEO, HEO & SSO have the highest success rates at 100%, with SO orbit having the lowest success rate at ~50%. Orbit SO has 0% success rate.
- LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit

Conclusions Cont....

- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. However for GTO we cannot distinguish this well as both positive landing rate and negative landing (unsuccessful mission) are both there here
- And finally the success rate since 2013 kept increasing till 2020.

Thank you!

